

A comparative analysis of Von-Heijne, SVM and Perceptron Models for the detection of Signal Peptides.

Abstract

Motivation: This is the motivation

Results: These are the results.

Materials and Methods

Data Collection

Protein sequences were queried from UniprotKB, and collected into 2 distinct sets, positive and negative. For Positive sequences, we selected sequences having experimental evidence for the presence of a signal peptide (ft_signal_exp:*) For the negative set, proteins not annotated with a signal peptide and endowed with experimental evidence for subcellular localization in non-secretory compartments, namely nucleus (cc_scl_term_exp:SL-0191), peroxisome (cc_scl_term_exp:SL-0204), cell membrane / plasma membrane (cc_scl_term_exp:SL-0039), cytosol (cc_scl_term_exp:SL-0091), plastid (cc_scl_term_exp:SL-0209) or mitochondrion (cc_scl_term_exp:SL-0173).

Both sets were also limited to the taxa of eukaryotes and reviewed entries, and non-fragment nature.

Further selection on the positive entries was made to select proteins which had information about the cleavage site and a Signal Peptide with a length of at least 14bp.

For the negative set, additional information was added to the data representing the presence of a transmembrane helix, whose hydrophobicity profile mimics the one of a signal peptide.

Clustering

Clustering was performed using mmseqs2 [cit], using parameters =c 0.4 --min-seq-id 0.3 --cov-mode 0 --cluster-mode 1, giving clusters with less than 30% pairwise identity.